

From Pavement to Protection: The Aggie Micro-Mobility Map

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1. Research Background and Motivation

Domain. We study micromobility on the Texas A&M College Station campus: personal bikes, e-bikes and e-scooters, plus the Veo shared fleet. These devices now carry a large share of short campus trips, and they bring two problems with them: crashes and theft.

Why now. On September 3, 2026, the Texas A&M System Board of Regents warned that micromobility devices could be banned from campus if crashes do not come down within a year [1, 2]. Micromobility crashes on campus rose 37% from 2024 to 2025, to 52 incidents, about half with an injury [1]. Thefts climbed at the same time: University Police (UPD) reported more than 70 e-bike and scooter thefts between November 2025 and February 2026, and a 26% rise from 2024 to 2025 [3].

Why visualization. The evidence needed to answer the regents sits in separate places: incident reports in police alerts and the Clery crime log, rack and lane data in campus GIS layers, and ride activity in Veo's public feed. No one can see, in one view, whether thefts cluster where racks are scarce or how incidents line up with paths and riding times. A linked campus map and its companion views can turn those logs into targeted fixes rather than a blanket ban. The problem is also personal to us: a friend was injured in a spill on campus last semester, and one of us bought an e-scooter this semester.

2. Design Requirements

2.1 Who are the target users?

We design for three primary users and one potential audience. Their needs come from public sources today and will be validated directly (Section 2.2).

User	What they decide	What they need to see
Campus police (UPD)	Where and when to patrol; where to add cameras	Theft density by place and time, to deploy officers and monitoring
Campus planners (Transportation Services)	Which racks, lanes and paths to fund first	Rack capacity, lane and path condition, and parking pressure by area
Student commuters	Which route to take and where to park	Lower-risk routes and parking with space near their buildings
Board of Regents (potential)	Whether to restrict or ban micromobility	Evidence on whether problems are falling and which fixes target them

2.2 How do we interact with target users to understand the design requirements?

We work in two stages. No direct user contact has happened yet; the stage-two instruments are drafted.

- **Stage 1, secondary sources (done).** First-pass needs drawn from public statements at the September 3 regents meeting [1, 2], UPD theft releases [3, 4] and Transportation Services interviews in *The Battalion* [5].
- **Interviews (next).** One 30-minute interview each with a UPD officer and a Transportation Services planner, via the public contact bikes@tamu.edu: what they decide each week, what data they use now, and what they cannot see.
- **Student survey (next).** A 5-minute anonymous survey of riders and walkers on routes, parking habits, near misses, theft experience and support for fixes; target 200 responses, at least 60 from people who only walk. It also measures how many crashes and thefts go unreported.
- **Regents and prototype testing.** The recorded regents meeting stands in for an interview. Each user group then gets one think-aloud session on the working prototype, and each requirement is marked confirmed, changed or dropped.

2.3 What are the design requirements?

ID	The dashboard must let users...	Main users
R1	Pinpoint theft hotspots, to direct patrols and camera placement	Police, planners
R2	See when thefts happen by day of week and time window, to schedule enforcement	Police
R3	Audit rack capacity and security against theft locations, to find under-served or vulnerable sites	Planners, students
R4	Overlay paths, trails and shared-fleet activity, to judge route safety and where riders concentrate	Planners, students, regents
R5	Filter once and see every view update (linked views), with incident details on demand	All
R6	See the source and coverage of every number, so reporting changes are not mistaken for crime changes (UPD's public list starts in Nov 2025)	All, esp. regents

2.4 What is the data?

Dataset	Source	Size	Geometry	Status
Bike theft incidents	UPD crime alerts and Clery crime log	125 thefts, 121 placed in zones	Table + points (zone level)	Ready
Bike rack assets	TAMU Transportation Services rack inventory [6]	316 racks with capacity	Points + table	Ready
Campus zones / risk sectors	Built by the team from rack clusters	91 zones; 150 m grid planned	Points now, polygons planned	Partly ready
Trails and pathways	City of College Station GIS trails layer [7]	187 line features	Lines	In review
Shared micromobility feed	Veo public GBFS feed [8]	~1,250 vehicles per snapshot	Table + points over time	Polling starting

Known limits. Theft times are reported as windows (median about 7 hours), so hour-level views suit Veo usage better than thefts. Theft records merge bikes, e-bikes and e-scooters and do not record lock type. The Veo feed shows only the current moment, so we poll it to build history, and it covers shared bikes only. Crash data exists only as yearly totals [1].

2.5 What are the data visualization tasks?

Task	Action → target [9, 10]	Req.	Visual encoding	Readiness
T1. Delineate and prioritize high-risk theft zones	Locate → spatial clusters	R1	Bubble map by zone; grid choropleth once the 150 m grid exists	Ready
T2. Audit rack proximity to high-theft zones	Compare → capacity vs. thefts	R3	Layered map, thefts and racks sized by count and capacity	Ready
T3. Analyze weekly and daily patterns	Identify → peaks	R2	Hour × weekday heatmap	Caveat: time windows
T4. Track trends over time and by semester	Discover → trend	R2, R6	Annotated line chart, marked by source	Caveat: coverage from Nov 2025
T5. Compare by vehicle type and security	Compare → distributions	R3	Grouped or 100% stacked bars	Needs data
T6. Rank campus sectors and facilities	Identify → extremes	R1, R3	Sorted horizontal bar chart	Ready
T7. Retrieve incident details on demand	Query → one item	R5	Linked views with brushing and tooltips	Ready

The views follow “overview first, zoom and filter, then details on demand” [11]: the map gives the overview, filters and brushing narrow it, and tooltips show single incidents.

3. Related Work

3.1 What are the existing systems and research work?

System	What it does	Gap for our users
micromodes.org, UNC Highway Safety Research Center [12]	National dashboard of e-scooter fatalities	Macro scale; no campus infrastructure or theft
Brown and Howell equity dashboard [13]	Equity requirements across 239 U.S. shared micromobility programs	Policy level; no incident or infrastructure data
Ride Report and city dashboards [14]	Fleet monitoring and compliance from operator feeds (GBFS, MDS)	City-wide fleets; no theft, personal devices or campus detail

System	What it does	Gap for our users
MicroVis, Victoria Univ. of Wellington [15]	Analyst dashboard of linked widgets for e-scooter trips	Usage only; no theft or safety layer
Wood, Slingsby and Dykes [16]	Requirements-driven flow maps of London bicycle-hire use	Usage only; city scale

3.2 What is the potential novelty of our system?

Users today see three separate pictures: campus maps for infrastructure, police logs for security and the Veo app for mobility. Our system joins them.

1. **A self-built, multi-source dataset.** We scrape, clean and join UPD theft records with polled Veo availability (a history that does not exist publicly) and public campus and city GIS layers.
2. **Campus-scale geography.** A 150 m grid built for a dense university campus replaces city districts, so racks, buildings and paths fall in separate cells.
3. **Coverage-aware views.** Each count carries its source, so a change in reporting is shown as such (R6). With about 120 located thefts, the system surfaces candidate relationships to investigate; it does not claim to prove causes.

4. Development Plans

4.1 What are the task assignments of our team members?

Member	Role	Responsibilities
Dushyant Betala	Backend	WP1–WP2: data mining from city and campus GIS and supplementary sources; cleaning, structuring, normalizing
Pratyush Kumar	Full-stack	WP1–WP3: Veo and UPD theft data mining and preparation; spatial joins; project architecture and deployment
Avneesh Dubey	Frontend	WP4: dashboard design and visual encoding decisions; D3.js view development
Shanmukh Nutulapati	Frontend	WP4: D3.js views and interactions; integrating backend data into the frontend

WP1 data mining · WP2 cleaning, deduplication, geocoding and reprojection to one schema · WP3 spatial joins and GeoJSON/JSON export · WP4 five to six coordinated D3.js views with linked brushing, filtering, zoom and tooltips, deployed on GitHub Pages.

4.2 What is the timeline?

Data work runs through October; the D3 views finish by Nov 17

Group 9 work plan by week, 2026, read from the project Gantt

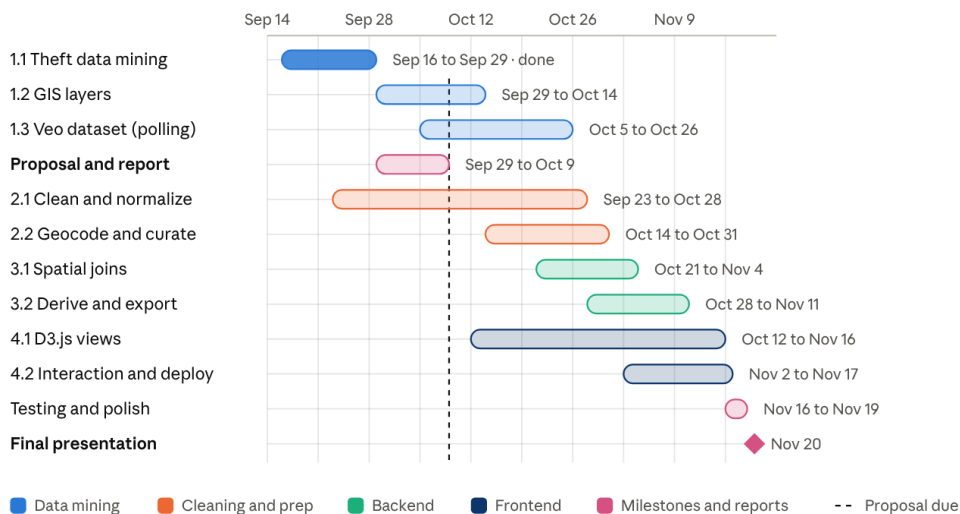


Figure 1. Work plan by week. Theft data mining is done; key milestones are this proposal (Oct 9), end of testing (Nov 19) and the final presentation (Nov 20).

5. Reflection on Using AI

The course asks every team to use AI and reflect on it. We used AI assistants (Claude and Claude Code) as a support tool; scope, data strategy, user analysis and every design decision came from the team. Each member who used AI uploads their interaction logs with this proposal.

5.1 Where we used AI, and what stayed ours

Area	What the AI did	What the team decided	How we checked it
Writing	Refined wording in the slides, task descriptions and this report	Scope, task breakdown, roles and every claim	Read against the deck and sources; cut overstatements
Research support	Found public data sources (Veo GBFS, campus and city GIS services) and related systems	Which sources to use and how to combine them	Opened each source; ran live queries to confirm counts
Visual prototyping	Drafted layouts such as the timeline chart and an early dashboard mock-up	Dates, milestones, encodings and styling	Compared against our Gantt and data; fixed label collisions
Engineering scaffolding	Drafted the repository structure, CI templates and architecture document with Claude Code	Architecture, data pipeline and deployment choices	Code review; commit trailers mark AI-assisted commits
Instrument drafting	Drafted the student survey and interview questions	Questions kept, wording, routing and target users	Pilot with five people before launch

5.2 What worked

- **Speed on routine work.** Boilerplate code, formatting and first drafts took minutes, leaving more time for data cleaning and design decisions.
- **Breadth of search.** The AI surfaced queryable data services we had not found, such as the campus bike-map layers for racks, lanes and dismount zones.
- **A second reader.** Asking it to check a draft against the rubric caught missing items and inconsistent numbers.

5.3 Limits we saw, and what we changed

- **Planned work described as built.** AI-drafted documents described planned features as finished. We now mark every item as built, in progress or planned.
- **Overclaiming from small data.** Drafts said the dashboard would “reveal correlations.” With about 120 located thefts we cannot claim that, so we reworded to “explore candidate relationships.”
- **Unverified figures.** Numbers from search snippets were not trusted until we opened the original article or queried the data ourselves.

5.4 How we keep AI use accountable

- Every AI suggestion is reviewed and edited by a team member before it is used.
- The data export is reproducible: re-running the pipeline produces an identical file, and the theft-to-zone mapping is reviewed by hand.
- AI-assisted commits carry co-author trailers, and each member's interaction logs and reflection are submitted with the proposal.

References

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